

Integration Techniques: 100 Practice Problems

Evaluate each indefinite integral. Include the constant of integration C in every answer. Click a blue integral to show its answer; click it again to hide the answer.

1. $\int 2x(x^2 + 3)^5 dx$ [show/hide answer]

2. $\int \frac{dx}{\sqrt{9 - x^2}}$ [show/hide answer]

3. $\int xe^x dx$ [show/hide answer]

4. $\int \sin^3 x \cos^2 x dx$ [show/hide answer]

5. $\int \frac{3x^2}{x^3 + 4} dx$ [show/hide answer]

6. $\int \sqrt{16 - x^2} dx$ [show/hide answer]

7. $\int x \cos x dx$ [show/hide answer]

8. $\int \sin^2 x \cos^2 x dx$ [show/hide answer]

9. $\int e^{5x-1} dx$ [show/hide answer]

10. $\int \frac{x^2}{\sqrt{25 - x^2}} dx$ [show/hide answer]

11. $\int x \ln x dx$ [show/hide answer]

12. $\int \cos^5 x \, dx$ [show/hide answer]
13. $\int x\sqrt{x^2 + 9} \, dx$ [show/hide answer]
14. $\int \frac{dx}{(x^2 + 4)^{3/2}}$ [show/hide answer]
15. $\int x^2 e^x \, dx$ [show/hide answer]
16. $\int \sin^4 x \, dx$ [show/hide answer]
17. $\int \cos(4x + 2) \, dx$ [show/hide answer]
18. $\int \sqrt{x^2 + 9} \, dx$ [show/hide answer]
19. $\int e^x \sin x \, dx$ [show/hide answer]
20. $\int \tan^3 x \sec^2 x \, dx$ [show/hide answer]
21. $\int \frac{(\ln x)^3}{x} \, dx$ [show/hide answer]
22. $\int \frac{x^2}{\sqrt{x^2 + 16}} \, dx$ [show/hide answer]
23. $\int \arctan x \, dx$ [show/hide answer]
24. $\int \sec^4 x \tan x \, dx$ [show/hide answer]
25. $\int \frac{x}{(x^2 + 1)^2} \, dx$ [show/hide answer]
26. $\int \frac{dx}{x^2 \sqrt{x^2 - 1}}$ [show/hide answer]

$$27. \int x \arctan x \, dx \text{ [show/hide answer]}$$

$$28. \int \tan^2 x \sec^4 x \, dx \text{ [show/hide answer]}$$

$$29. \int \sin x e^{\cos x} \, dx \text{ [show/hide answer]}$$

$$30. \int \frac{\sqrt{x^2 - 4}}{x} \, dx \text{ [show/hide answer]}$$

$$31. \int x^2 \ln x \, dx \text{ [show/hide answer]}$$

$$32. \int \sec^3 x \, dx \text{ [show/hide answer]}$$

$$33. \int \frac{1}{\sqrt{7 - 2x}} \, dx \text{ [show/hide answer]}$$

$$34. \int \frac{dx}{\sqrt{x^2 - 25}} \text{ [show/hide answer]}$$

$$35. \int e^{2x} \cos x \, dx \text{ [show/hide answer]}$$

$$36. \int \csc^3 x \, dx \text{ [show/hide answer]}$$

$$37. \int (6x - 1)\sqrt{3x^2 - x + 2} \, dx \text{ [show/hide answer]}$$

$$38. \int \frac{x^2}{(x^2 + 9)^{3/2}} \, dx \text{ [show/hide answer]}$$

$$39. \int \ln(1 + x) \, dx \text{ [show/hide answer]}$$

$$40. \int \sin(3x) \cos(5x) \, dx \text{ [show/hide answer]}$$

$$41. \int \frac{e^{\sqrt{x}}}{\sqrt{x}} \, dx \text{ [show/hide answer]}$$

42. $\int \frac{dx}{x^2 \sqrt{16 - x^2}}$ [show/hide answer]

43. $\int x \sin(2x) dx$ [show/hide answer]

44. $\int \cos^2(2x) dx$ [show/hide answer]

45. $\int \frac{\sec^2 x}{(1 + \tan x)^3} dx$ [show/hide answer]

46. $\int \frac{\sqrt{9 - x^2}}{x^2} dx$ [show/hide answer]

47. $\int x^3 \cos x dx$ [show/hide answer]

48. $\int \sin^2(4x) dx$ [show/hide answer]

49. $\int \frac{x^3}{x^4 + 5} dx$ [show/hide answer]

50. $\int \frac{dx}{(x^2 + 1)^{3/2}}$ [show/hide answer]

51. $\int x^2 \sin x dx$ [show/hide answer]

52. $\int \sin^3 x dx$ [show/hide answer]

53. $\int \frac{\cos x}{\sqrt{1 + \sin x}} dx$ [show/hide answer]

54. $\int \frac{x^2}{\sqrt{x^2 - 9}} dx$ [show/hide answer]

55. $\int \ln x dx$ [show/hide answer]

56. $\int \cos^3 x dx$ [show/hide answer]

57. $\int \frac{1}{x\sqrt{\ln x}} dx$ [show/hide answer]
58. $\int \frac{\sqrt{x^2 + 4}}{x^2} dx$ [show/hide answer]
59. $\int x e^{3x} dx$ [show/hide answer]
60. $\int \tan^4 x dx$ [show/hide answer]
61. $\int (2x + 1)e^{x^2+x} dx$ [show/hide answer]
62. $\int \frac{dx}{x\sqrt{x^2 + 9}}$ [show/hide answer]
63. $\int x \ln(2x) dx$ [show/hide answer]
64. $\int \sec^2 x \tan^3 x dx$ [show/hide answer]
65. $\int \frac{\ln(2x)}{x} dx$ [show/hide answer]
66. $\int \frac{dx}{(25 - x^2)^{3/2}}$ [show/hide answer]
67. $\int e^x \cos(2x) dx$ [show/hide answer]
68. $\int \sin^2 x \cos^4 x dx$ [show/hide answer]
69. $\int \frac{x^2}{(x^3 - 2)^4} dx$ [show/hide answer]
70. $\int \frac{x^2}{(x^2 - 4)^{3/2}} dx$ [show/hide answer]
71. $\int x^2 e^{2x} dx$ [show/hide answer]

$$72. \int \sin(2x) \sin(7x) dx \text{ [show/hide answer]}$$

$$73. \int \frac{\sin(3x)}{2 + \cos(3x)} dx \text{ [show/hide answer]}$$

$$74. \int \sqrt{x^2 - 16} dx \text{ [show/hide answer]}$$

$$75. \int x \arcsin x dx \text{ [show/hide answer]}$$

$$76. \int \cos(3x) \cos(6x) dx \text{ [show/hide answer]}$$

$$77. \int x^2 \sqrt{x^3 + 1} dx \text{ [show/hide answer]}$$

$$78. \int \frac{dx}{x^2 \sqrt{x^2 + 4}} \text{ [show/hide answer]}$$

$$79. \int \ln(x^2 + 1) dx \text{ [show/hide answer]}$$

$$80. \int \tan^3 x \sec^4 x dx \text{ [show/hide answer]}$$

$$81. \int \frac{e^x}{(1 + e^x)^2} dx \text{ [show/hide answer]}$$

$$82. \int \frac{x^4}{\sqrt{36 - x^2}} dx \text{ [show/hide answer]}$$

$$83. \int x^2 \cos(3x) dx \text{ [show/hide answer]}$$

$$84. \int \cot^2 x dx \text{ [show/hide answer]}$$

$$85. \int \frac{1}{x(1 + \ln x)^2} dx \text{ [show/hide answer]}$$

$$86. \int \frac{\sqrt{x^2 - 1}}{x^2} dx \text{ [show/hide answer]}$$

87. $\int x^3 e^{-x} dx$ [show/hide answer]

88. $\int \csc^2 x \cot^3 x dx$ [show/hide answer]

89. $\int \frac{4x^3}{\sqrt{x^4 + 6}} dx$ [show/hide answer]

90. $\int \frac{dx}{(x^2 + 25)^{5/2}}$ [show/hide answer]

91. $\int e^{2x} \sin(3x) dx$ [show/hide answer]

92. $\int \sin^5(2x) \cos(2x) dx$ [show/hide answer]

93. $\int \frac{\tan x}{\cos^2 x} dx$ [show/hide answer]

94. $\int \frac{x^2}{(x^2 + 1)^{5/2}} dx$ [show/hide answer]

95. $\int x \ln(x + 1) dx$ [show/hide answer]

96. $\int \cos^3(3x) \sin(3x) dx$ [show/hide answer]

97. $\int \frac{x}{\sqrt{4 - x^2}} dx$ [show/hide answer]

98. $\int \frac{dx}{x^3 \sqrt{x^2 - 9}}$ [show/hide answer]

99. $\int x^2 \arctan x dx$ [show/hide answer]

100. $\int \sec^5 x \tan x dx$ [show/hide answer]